

4. To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

```
GNU nano 8.7.1 skill3.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

#define MAX_INPUT 1024
#define MAX_TOKENS 100

int main()
{
    char input[MAX_INPUT];
    char *tokens[MAX_TOKENS];
    int tokenCount;

    printf("===== Mini Shell Parser =====\n");
    printf("Type 'exit' to quit.\n\n");

    while (1)
    {
        printf("MiniShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strlen(input) == 0)
        {
            printf("Empty command.\n");
            continue;
        }

        if (strcmp(input, "exit") == 0)
            break;

        tokenCount = 0;

        char *token = strtok(input, " \t");

        while (token != NULL && tokenCount < MAX_TOKENS - 1)
        {
            tokens[tokenCount++] = token;
            token = strtok(NULL, " \t");
        }
    }
}
```

```

GNU nano 8.7.1 skill3.c *
while (token != NULL && tokenCount < MAX_TOKENS - 1)
{
    tokens[tokenCount++] = token;
    token = strtok(NULL, " \t");
}

tokens[tokenCount] = NULL;

printf("\nParsed Tokens:\n");

for (int i = 0; i < tokenCount; i++)
{
    printf("Token %d : %s\n", i + 1, tokens[i]);
}

printf("\nCommand Structure:\n");

if (tokenCount > 0)
{
    printf("Command : %s\n", tokens[0]);

    if (tokenCount > 1)
    {
        printf("Arguments:\n");

        for (int i = 1; i < tokenCount; i++)
        {
            printf(" Arg %d : %s\n", i, tokens[i]);
        }
    }
    else
    {
        printf("No Arguments\n");
    }
}

printf("-----\n");

printf("Shell Closed.\n");

return 0;
}

```

```

thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ ls
2520090074_Skill  OSSkill  os  practial1  practical1.c  practical2.c  program1  program2
OSPractical      destination.txt  practcall.c  practical  practical2  practical4.c  program1.c  sample.ts
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ cd OSSkill
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ nano skill3.c
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ gcc skill3.c -o skill3
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~/OSSkill$ ./skill3
===== Mini Shell Parser =====
Type 'exit' to quit.

MiniShell> ls

Parsed Tokens:
Token 1 : ls

Command Structure:
Command : ls
No Arguments
-----
MiniShell> ls -l

Parsed Tokens:
Token 1 : ls
Token 2 : -l

Command Structure:
Command : ls
Arguments:
Arg 1 : -l
-----
MiniShell> |

```